



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PART TEST

07-01-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE-S, I(A), I & I(N)] : (SCORE-I)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,D	A,B	A,C,D	A,C,D	A,B,C	B,D	A,D	A,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	D	A	C	A				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.85 to 2.86	3.11	0.00	4.42 to 4.43	8.00	0.41 to 0.43		

PART-2 : CHEMISTRY

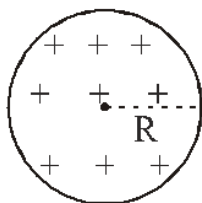
SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,C	B,C	A,B,D	B,C	A,B,C	A,B,C,D	A,C,D	B,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	D	C	C	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	0.60	40.00 to 40.20	6.00	1.50	0.00	825.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,C	A,B,C,D	A,C	A,B	A,B,D	B,D	B	B
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	D	A	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	0.56	0.25	1.10	0.17	13.00	13.60		

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A,B,D)



$$V_{\text{center}} = \int_0^R \frac{\frac{q}{\pi R^2} 2\pi r dr}{4\pi \epsilon_0 r}$$

$$V_{\text{center}} = \frac{q}{2\pi \epsilon_0 R}$$

$$\text{self PE of disc} = \int v dq$$

$$= \int_0^R \frac{\sigma r}{\pi \epsilon_0} 2\pi r dr \sigma = \frac{2\sigma^2}{\epsilon_0} \int_0^R r^2 dr = \frac{2\sigma^2}{\epsilon_0} \frac{R^3}{3}$$

$$= \frac{2R^3}{3\epsilon_0} \frac{q^2}{\pi^2 R^4} = \frac{2q^2}{3\pi^2 \epsilon_0 R}$$

$$\text{self PE of solid sphere} = \frac{3q^2}{20\pi \epsilon_0 R}$$

$$W_{\text{elec}} = -\Delta U = \frac{q^2}{R \epsilon_0} \left[\frac{2}{3\pi^2} - \frac{3}{20\pi} \right]$$

$$\text{ratio} = \frac{40}{9\pi}$$

ratio of potentials at center

$$= \frac{V_{\text{disc}}}{V_{\text{sphere}}} = \frac{\frac{q}{2\pi \epsilon_0 R}}{\frac{3}{2} \left(\frac{q}{4\pi \epsilon_0 R} \right)} = \frac{1}{2\pi} \times \frac{8\pi}{3} = \frac{4}{3}$$

2. Ans (A,B)

The angle between dipole moment vector and radius vector γ is given by

$$\cos \gamma = \hat{j} \cdot [(\sin \theta \cos \phi) \hat{i} + (\sin \theta \sin \phi) \hat{j} + (\cos \theta) \hat{k}]$$

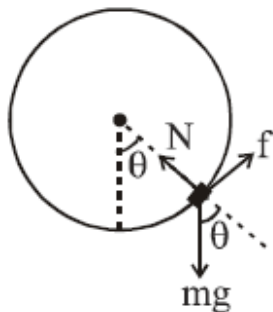
$$= \sin \theta \sin \phi$$

$$\therefore \text{Potential, } V = \frac{kp \cos \gamma}{r^2} = \frac{p \sin \theta \sin \phi}{4\pi\epsilon_0 r^2}$$

and magnitude of electric field

$$E = \frac{kp}{r^3} \sqrt{1 + 3\cos^2 \gamma} = \frac{p}{4\pi\epsilon_0 r^3} \sqrt{1 + 3\sin^2 \theta \sin^2 \phi}$$

3. Ans (A,C,D)



$$N = mg \cos \theta + m\omega^2 R \quad \dots\dots\dots(i)$$

$$f = mg \sin \theta \quad \dots\dots\dots(ii)$$

$$mg \sin \theta \leq \mu mg \cos \theta + \mu m\omega^2 R$$

$$\omega \geq \sqrt{\frac{g \sin \theta - \mu g \cos \theta}{\mu R}}$$

$$\text{as } [\sin \theta - \mu \cos \theta]_{\max} = \sqrt{1 + \mu^2}$$

$$\omega \geq \sqrt{\frac{g}{\mu R}} \sqrt{1 + \mu^2}$$

$$\text{from (i) \& (ii) } f_{\max} = mg \quad (\theta = 90^\circ)$$

$$N_{\min} = -mg + m\omega^2 R \quad (\theta = 90^\circ)$$

4. Ans (A,C,D)

Suppose that the perigee and apogee of a satellite orbit are at height h and H above the Earth. The corresponding apsidal distances are therefore $R + h$ and $R + H$ respectively, where R is the Earth's radius. Then

$$(R + h) + (R + H) = 2a,$$

where a is the semi-major axis of the orbit. The parameter a is therefore given by

$$a = R + \frac{1}{2}(h + H) = 6400 + \frac{1}{2}(181 + 327) = 6654 \text{ km}$$

The period τ of the orbit can now be found from the period formula

$$\tau^2 = \frac{4\pi^2 a^3}{MG} = \frac{4 \times 3.14^2 \times 6654^3 \times 10^9}{10 \times 6400^2 \times 10^6} \Rightarrow \tau = 53265$$

where M is the mass of the Earth.

Let V be the speed of the satellite at the perigee. Then, from the energy conservation equation and the E-formula.

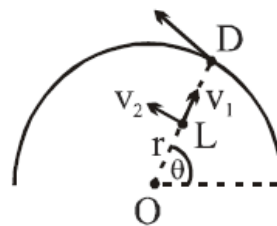
$$\frac{1}{2} V^2 - \frac{MG}{R + h} = \frac{MG}{2a}$$

so that the speed at the perigee is given by

$$V^2 = MG \left(\frac{2}{R + h} - \frac{1}{a} \right) \Rightarrow V \cong 8 \text{ km / sec}$$

5. Ans (A,B,C)

v_1 & v_2 are radial and tangential components of lion's velocity.



$$v_1^2 + v_2^2 = v^2 \quad \dots\dots(1)$$

$$OL = r \quad \dots\dots(2)$$

$$\frac{d\theta}{dt} = \frac{u}{L} = \frac{v_2}{r} \quad \dots\dots(3)$$

$$v_1 = \frac{dr}{dt} \quad \dots\dots(4)$$

$$\text{from these equations } \Rightarrow \frac{dr}{dt} = \frac{u}{a} \sqrt{\frac{v^2 a^2}{u^2} - r^2}$$

$$\int_0^r \frac{dr}{\sqrt{\frac{v^2 a^2}{u^2} - r^2}} = \int_0^t \frac{u}{a} dt$$

$$r = \frac{va}{u} \sin \frac{ut}{a}$$

$\therefore$ separation between lion and Dinanath is $a-r$

$$= a - \frac{va}{u} \sin \frac{ut}{a}$$

Chase ends at $r = a$

$$\Rightarrow t = \frac{a}{u} \sin^{-1} \frac{u}{v}$$

path of Lion : $x = r \cos \theta$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$x^2 + y^2 = \frac{yva}{u}$$

$$x^2 + \left(y - \frac{va}{2u}\right)^2 = \left(\frac{va}{2u}\right)^2$$

6. **Ans (B,D)**

Initial velocity of particle w.r.t ground

$$= v_2 \hat{i} + 25\hat{j} + v_1 \hat{k}$$

Initial velocity of particle w.r.t platform

$$= v_2 \hat{i} + 25\hat{j} + v_1 \hat{k}$$

Acceleration of particle w.r.t. platform

$$= \vec{a} = -2\hat{i} - 12.5\hat{j}$$

$$\text{w.r.t platform: } S_x = u_x t + \frac{1}{2} a_x t^2$$

$$S_x = v_x t - t^2; S_y = 25t + \frac{1}{2}(-12.5)t^2 = 0$$

$$t = 4 \text{ sec.}$$

$$S_z = v_1 t$$

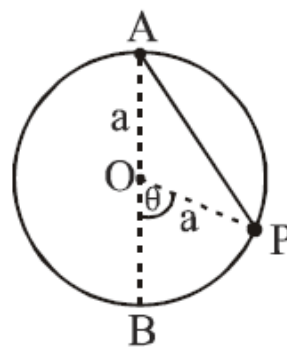
$$8 \leq S_x \leq 16; 16 \leq S_z \leq 24$$

$$8 \leq v_2 t - t^2 \leq 16; 16 \leq v_1 t \leq 24$$

$$8 \leq v_2 (4) - 16 \leq 16; 16 \leq 4v_1 \leq 24$$

$$24 \leq 4v_2 \leq 32; 4 \leq v_1 \leq 6$$

7. **Ans (A,D)**



Since the wire is smooth the constraint force that it exerts on the particle does no work.

Thus energy conservation holds in its standard form. Let θ be the angle between the radius OP and the downwards vertical, as shown in figure.

The length of the spring is $2a \cos \frac{1}{2} \theta$ and its potential energy is therefore.

$$\frac{1}{2} \alpha \left(2a \cos \frac{1}{2} \theta - \frac{3}{2} a \right)^2 = \frac{\alpha a^2}{8} \left(4 \cos \frac{1}{2} \theta - 3 \right)^2$$

The gravitational potential energy of the particle is $-mg a \cos \theta$. The total potential energy of the system is therefore

$$V = \frac{\alpha a^2}{8} \left(4 \cos \frac{1}{2} \theta - 3 \right)^2 - m g a \cos \theta.$$

on differentiating, we find that

$$V' = \frac{3}{2} \alpha a^2 \sin \frac{1}{2} \theta + (m g a - \alpha a^2) \sin \theta$$

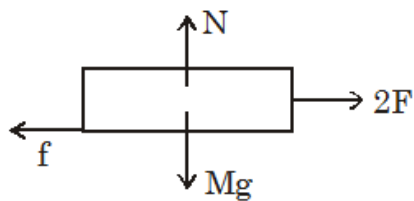
and

$$V'' = \frac{3}{4} \alpha a^2 \cos \frac{1}{2} \theta + (m g a - \alpha a^2) \cos \theta$$

Hence

$$V'(0) = 0 \text{ and } V''(0) = m g a - \frac{1}{4} \alpha a^2.$$

8. Ans (A,C)



$$2F = f \leq \mu_s Mg$$

$$2Kt^2 \leq \mu_s Mg$$

$$t^2 \leq \frac{\mu_s Mg}{2K}$$

$$t^2 = \frac{0.5 \times 150 \times 10}{2 \times 60}$$

$$t \leq 2.5 \text{ sec}$$

for $t > 2.5 \text{ sec}$

$$2Kt^2 - f_k = Ma$$

$$2Kt^2 - \mu_k Mg = Ma$$

$$\frac{2K}{M}t^2 - \mu_k g = a$$

$$t = 5 \text{ sec}$$

$$a = 0.4 [2t^2 - 10]$$

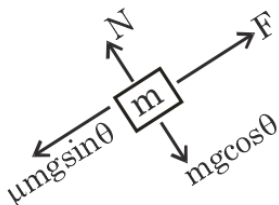
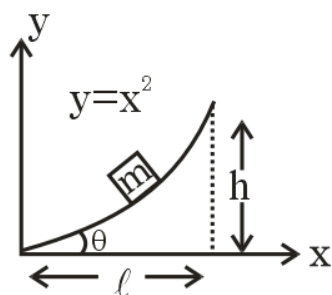
$$a = 0.4 [50 - 10]$$

$$a = 16 \text{ m/s}^2$$

PART-1 : PHYSICS

SECTION-I (ii)

9. Ans (D)



$$w_f = \mu mg \ell$$

$$w_f = 10 \text{ J}$$

$$\tan \theta = \frac{dy}{dx}$$

$$\tan \theta = 2x$$

$$\tan \theta = 1 \text{ at } x = \frac{1}{2}$$

$$\theta = 45^\circ$$

$$N = mg \cos \theta$$

$$N = 2 \times 10 \times \frac{1}{\sqrt{2}} = 10\sqrt{2} \text{ N}$$

$$W_F = \mu mg \ell + mgh$$

$$W_F = 0.5 \times 20 \times 1 + 20 \times 1$$

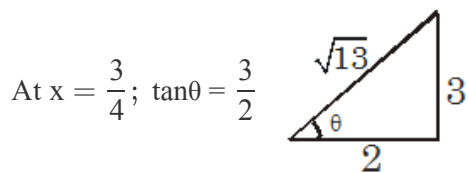
$$W_F = 30 \text{ J}$$

At any point : $N = mg \cos \theta$

$$f = \mu mg \cos \theta$$

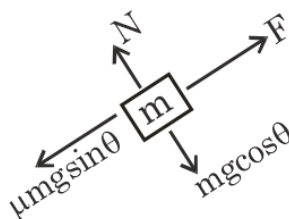
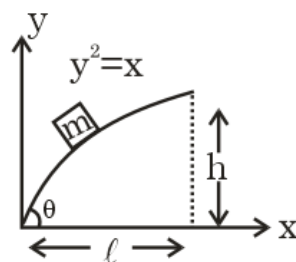
$$R = \sqrt{N^2 + f^2}$$

$$R = mg \cos \theta \sqrt{1 + \mu^2}$$



$$R = 20 \times \frac{2}{\sqrt{13}} \times \sqrt{1 + \frac{1}{4}} = 20\sqrt{\frac{5}{13}} \text{ N}$$

10. Ans (A)



$$w_f = \mu mg \ell$$

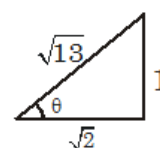
$$w_f = 10 \text{ J}$$

$$\tan \theta = \frac{dy}{dx} \quad 2y \frac{dy}{dx} = 1$$

$$\tan \theta = \frac{1}{2y}$$

$$\tan \theta = \frac{1}{2\sqrt{x}}$$

$$\tan \theta = \frac{1}{\sqrt{2}}, \text{ at } x = \frac{1}{2}$$



$$N = mg \cos \theta$$

$$N = 2 \times 10 \times \sqrt{\frac{2}{3}} = 20\sqrt{\frac{2}{3}} \text{ N}$$

$$w_f = \mu mg \ell + mgh$$

$$w_f = 0.5 \times 20 \times 1 + 20 \times 1$$

$$w_f = 30 \text{ J}$$

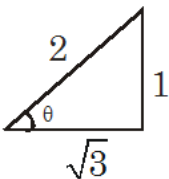
$$\text{At any point : } N = mg \cos \theta$$

$$f = \mu mg \cos \theta$$

$$R = \sqrt{N^2 + f^2}$$

$$R = mg \cos \theta \sqrt{1 + \mu^2}$$

At $x = \frac{3}{4}$; $\tan \theta = \frac{1}{\sqrt{3}}$



$$R = 20 \times \frac{\sqrt{3}}{2} \times \sqrt{1 + \frac{1}{4}} = 5\sqrt{15} \text{ N}$$

11. **Ans (C)**

$$(I) E_B = \frac{\lambda}{2\pi\epsilon_0 R} = \frac{Q}{2\pi\epsilon_0 R^2} \therefore N = \frac{1}{2\pi}$$

(II) Take a ring element at azimuthal angle θ and angular thickness $d\theta$

$$\text{Area } dA = 2\pi R^2 \sin \theta d\theta$$

$$E = \int_0^{\pi/2} \frac{\sigma dA (R \cos \theta)}{4\pi\epsilon_0 R^3} = \frac{Q}{8\pi\epsilon_0 R^2}$$

$$N = \frac{1}{8\pi}$$

$$(III) E = 0$$

$$N = 0$$

$$(IV) E = 2 \left[\frac{Q}{4\pi\epsilon_0 R^2} \right] \frac{\sin(\pi/2)}{(\pi/2)}$$

$$N = \frac{1}{\pi^2}$$

12. **Ans (A)**

$$(I) \left| \Delta V_{AB} \right| = \frac{\lambda}{2\pi\epsilon_0} \ln 2 = \frac{Q \ln 2}{2\pi\epsilon_0 R}$$

$$N = \frac{\ln 2}{2\pi}$$

(II) $\left| \Delta V_{AB} \right| = 0$ as electric field is perpendicular to equatorial plane

$$(III) \left| \Delta V_{AB} \right| = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{R} - \frac{1}{R\sqrt{2}} \right) \Rightarrow N = \frac{\sqrt{2}-1}{4\sqrt{2}\pi}$$

(IV) $\left| \Delta V_{AB} \right| = 0$ as positive and negative charge is equidistant from A as well as B

PART-1 : PHYSICS

SECTION-II

1. **Ans (2.85 to 2.86)**

Newton second Law:

$$\frac{d}{dt} (mv) = mg$$

$$\frac{d}{dt} \left(\frac{4}{3} \pi r^3 \rho v \right) = \frac{4\pi r^3 \rho g}{3}$$

$$\frac{d}{dt} (r^3 v) = r^3 g \dots\dots(1)$$

given :

$$\frac{dm}{dt} = Kvr^2 \quad (K = \text{prop const tan t})$$

$$\frac{d}{dt} \left(\frac{4\pi r^3 \rho}{3} \right) = Kvr^2$$

$$\frac{dr}{dt} = kv \quad \dots\dots(1) \quad (k = \text{New Prop constant})$$

$$\text{Now } \frac{d}{dt} (r^3 v) = \frac{d}{dr} (r^3 v) \frac{dr}{dt}$$

$$r^3 g = kv \left(3r^2 v + \frac{r^3 dv}{dr} \right) \quad [\text{from } \dots\dots(1)]$$

$$r^3 g = 3r^2 v^2 k + r^3 kv \frac{dv}{dr}$$

$$\frac{rv dv}{dr} + 3v^2 = \frac{gr}{k}$$

$$\frac{d}{dr} (v^2) + \frac{6v^2}{r} = \frac{2g}{k}$$

$$v^2 = \frac{2gr}{7k} + \frac{C}{r^6}$$

$$\text{at } t = 0 ; r \rightarrow 0 \text{ \& } v = 0 \therefore C = 0$$

$$\therefore v^2 = \frac{2gr}{7k}$$

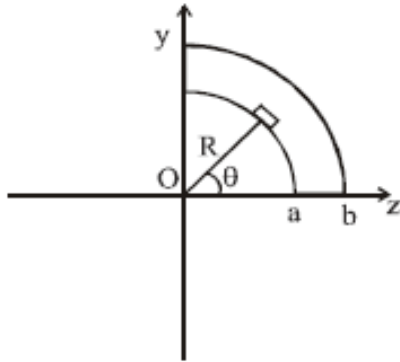
$$\frac{2v dv}{dt} = \frac{2g}{7k} \frac{dr}{dt}$$

$$\frac{2v dv}{dt} = \frac{2g}{7k} kv \Rightarrow \frac{dv}{dt} = \frac{2g}{7}$$

$$\frac{dv}{dt} = 2.85 \text{ m/s}^2$$

2. Ans (3.11)

Let the uniform mass density = σ



$$M = \int \sigma r dr d\theta$$

$$= \sigma \int_a^b r dr \int_0^{\sqrt{1/2}} d\theta = \frac{\pi}{4} (b^2 - a^2)$$

$$\vec{R} = \frac{1}{M} \int \vec{r} dm = \frac{\sigma}{M} \int (r \cos \theta \hat{i} + r \sin \theta \hat{j}) r dr d\theta$$

$$= \frac{\sigma}{M} \int_a^b r^2 dr \left[\int_0^{\pi/2} \cos \theta d\theta \hat{i} + \int_0^{\pi/2} \sin \theta d\theta \hat{j} \right]$$

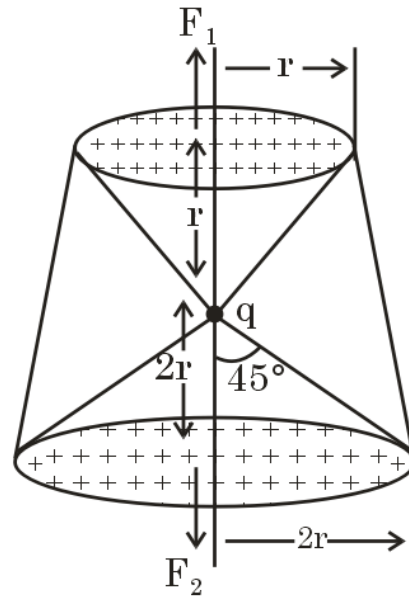
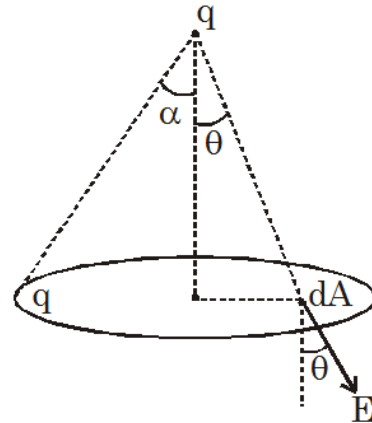
$$= \frac{4}{3\pi} \frac{(b^3 - a^3)}{b^2 - a^2} (\hat{i} + \hat{j})$$

$$\vec{R} = \frac{4}{3\pi} \frac{((2a)^3 - a^3)}{((2a)^2 - a^2)} (\hat{i} + \hat{j})$$

$$\vec{R} = \frac{4}{3\pi} \times \frac{7}{3} (\hat{i} + \hat{j})$$

$$K = \frac{28}{9} = 3.11 \text{ Ans.}$$

3. Ans (0.00)



Element of charge $dQ = \sigma dA$

E = electric field by q at dA

$$\text{force : } F = \int \sigma dA E \cos \theta$$

$$\text{on disc} = \sigma \int \vec{E} \cdot d\vec{A}$$

$$= \sigma \phi$$

$$= \sigma \frac{q}{2\epsilon_0} (1 - \cos \alpha) \downarrow$$

In our problem :

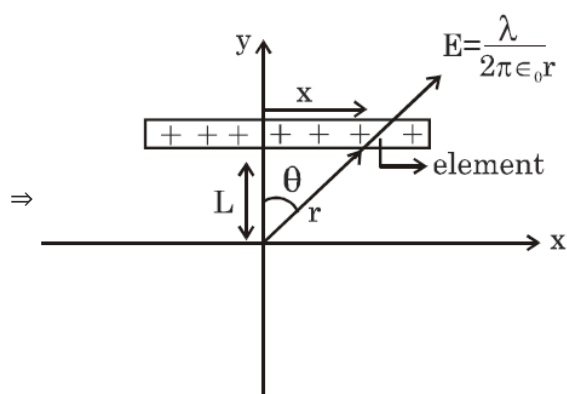
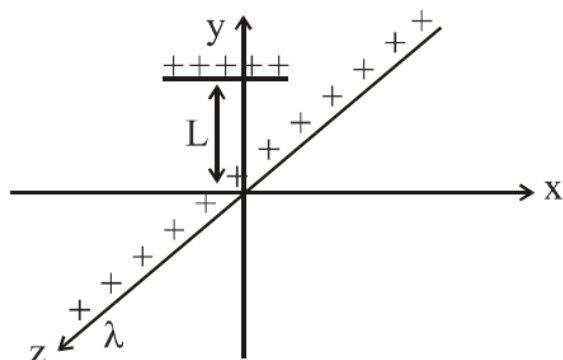
$$F_1 = \frac{\sigma q}{2\epsilon_0} (1 - \cos 45^\circ)$$

$$F_2 = \frac{\sigma q}{2\epsilon_0} (1 - \cos 45^\circ)$$

$$F_1 = F_2$$

$$F_{\text{Net}} \text{ by frustum on } q = 0$$

4. Ans (4.42 to 4.43)



$$\text{Force on element } dF = (2\lambda dx) \frac{\lambda}{2\pi\epsilon_0 r}$$

$$F_{\text{net}} = \int dF \cos \theta$$

$$F_{\text{net}} = \int \frac{\lambda^2 dx}{\pi\epsilon_0 r} \left(\frac{L}{r} \right)$$

$$F_{\text{net}} = \frac{\lambda^2 L}{\pi\epsilon_0} \int_{-L}^{+L} \frac{dx}{x^2 + L^2}$$

$$F_{\text{net}} = \frac{\lambda^2 L}{\pi\epsilon_0 L} \left[\tan^{-1} \frac{x}{L} \right]_{-L}^{+L}$$

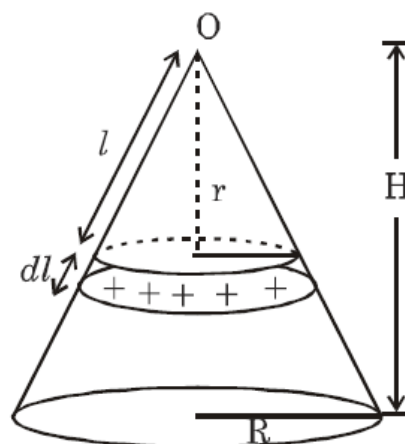
$$F_{\text{net}} = \frac{\lambda^2}{\pi\epsilon_0} \left[\frac{\pi}{2} \right] = \frac{\lambda^2}{2\epsilon_0}$$

$$F_{\text{net}} = \frac{8.85 \times 8.85}{2 \times 8.85 \times 10^{-12}}$$

$$F_{\text{net}} = 4.425 \times 10^{12} \text{ N}$$

5. Ans (8.00)

$$dq = \sigma 2\pi r d\ell$$



$$dV = \frac{dq}{4\pi\epsilon_0 \ell}$$

$$dV = \frac{\sigma 2\pi r d\ell}{4\pi\epsilon_0 \ell}$$

$$V = \frac{\sigma}{2\epsilon_0} \frac{R}{L} \int_0^L d\ell$$

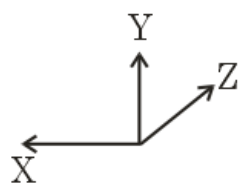
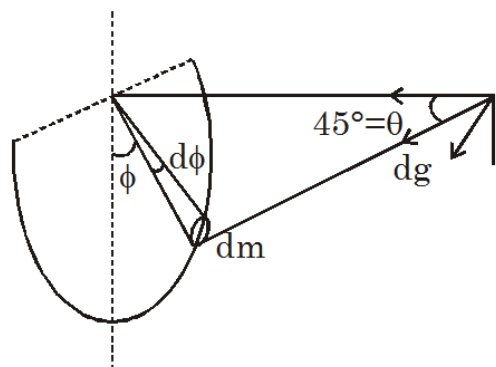
$$V = \frac{\sigma R}{2\epsilon_0}$$

$$U = qV$$

$$U = 1.6 \times 10^{-19} \times \frac{8.85 \times 10^{-10} \times 0.1}{2 \times 8.85 \times 10^{-12}}$$

$$U = 8 \times 10^{-19} \text{ J}$$

6. Ans (0.41 to 0.43)



$$g_z = \int_{-\pi/2}^{\pi/2} dg \sin \theta \sin \phi = 0$$

$$g_x = \int_{-\pi/2}^{\pi/2} dg \cos \theta = \frac{GMx}{[R^2 + x^2]^{3/2}} \quad [x = R]$$

$$g_y = \int_{-\pi/2}^{\pi/2} dg \sin \theta \cos \phi = \frac{Gdm}{R^2 + x^2} \sin \theta \cos \phi = \frac{2G\lambda R \sin \theta}{R^2 + x^2}$$

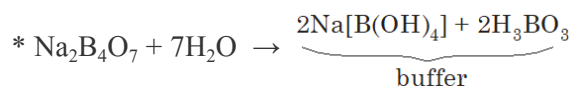
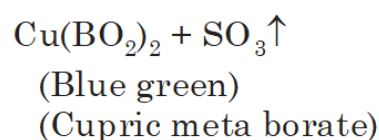
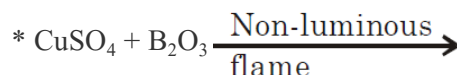
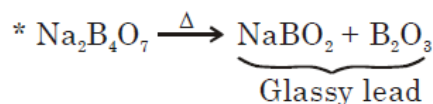
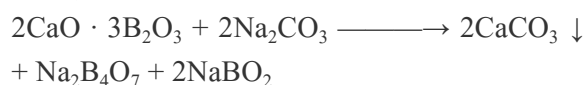
$$dm = \lambda R d\phi, \sin \theta = \frac{R}{\sqrt{R^2 + x^2}}$$

$$\therefore g_{\text{net}} = \sqrt{g_x^2 + g_y^2}$$

PART-2 : CHEMISTRY

SECTION-I (i)

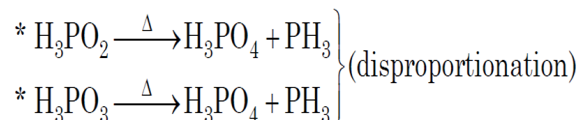
1. Ans (B,C)



* In borax, 2B are sp^2 hybridised
and 2B are sp^3 hybridised

2. Ans (B,C)

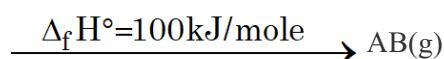
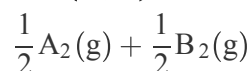
Order of acidity of H_3PO_2 , H_3PO_3 , H_3PO_4 is same as order of reducing nature.



Among $\text{H}_4\text{P}_2\text{O}_6$, $\text{H}_4\text{P}_2\text{O}_7$, $\text{H}_4\text{P}_2\text{O}_8$, only $\text{H}_4\text{P}_2\text{O}_7$ contain P-O-P linkage.

In oxo acids of P, there is $3d\pi-2p\pi$ overlapping.

4. Ans (B,C)

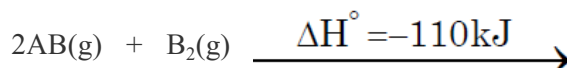


oxidation of A_2 by B_2 into AB_2



$2\text{AB}_2(\text{g}) \rightleftharpoons$ endothermic

oxidation of AB by B_2 into $\text{AB}_2(\text{g})$



$2\text{AB}_2 - (3) \Rightarrow$ exothermic

5. Ans (A,B,C)

Liquid $\xrightarrow[546\text{K}, 1\text{ atm}]{\Delta_r H = 273\text{L-atm/mole}}$ Vapour

$$\therefore \Delta H = 273 \times 5 = 1365 \text{ L-atm} = Q_p$$

$$\begin{aligned} \text{Workdone} &= -P_{\text{ext.}} (V_2 - V_1) \\ &\simeq -1 \times 22.4 \times 5 \times 2 \\ &= -224 \text{ L-atm} \end{aligned}$$

For this phase transition

$$\Delta S = \frac{\Delta H}{T} = \frac{273 \times 5}{273 \times 2} = 2.5 \text{ L-atm/K}$$

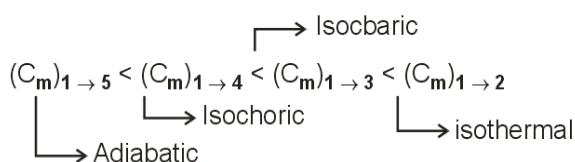
$$\Delta U = Q_p + W = 1365 - 224 = 1141 \text{ L-atm}$$

6. Ans (A,B,C,D)

$$dS = \frac{dQ_{\text{rev}}}{T} = \frac{nc_m dT}{T}$$

$$\left. \frac{dT}{dS} \right| = \frac{1}{nC_m}$$

at $T = T_1$



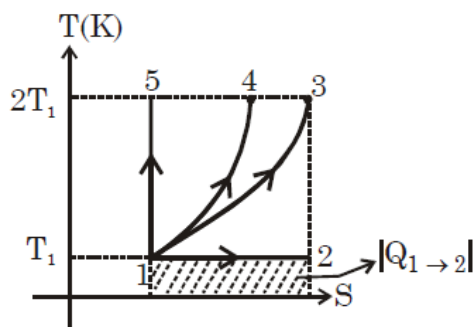
$\Rightarrow 1 \rightarrow 4$: Isochoric expansion process

$$\Delta S_{1 \rightarrow 4} = 2.303 nC_m \log \frac{2T_1}{T_1}$$

$$2.303 \times 2 \times \frac{5}{2} \times R \times 0.3 = 28.672 \text{ J/K}$$

$1 \rightarrow 3$: Isobaric expansion process

$$\Delta H = nC_{p,m} \Delta T = 2 \times \frac{7}{2} R (2T_1 - T_1) = 7 RT_1$$



$$\therefore dQ_{\text{rev}} = T dS$$

$$Q = \int_i^f T dS \Rightarrow |Q| = \text{area bounded by } T - S \text{ curve}$$

$$Q_{1 \rightarrow 5} < Q_{1 \rightarrow 2} < Q_{1 \rightarrow 3}$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (0.60)

$$Q + W = \Delta U$$

$$\Rightarrow y - \frac{y}{2} = \Delta U = nC_{v,m} \Delta T$$

$$y = 2n C_{v,m} \Delta T$$

$$\text{using } Q = nC_m \Delta T$$

$$\Rightarrow nC_m \Delta T = 2nC_{v,m} \Delta T$$

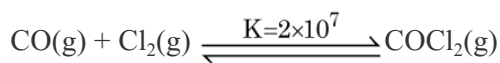
$$\Rightarrow C_m = 2C_{v,m} = 2 \times \frac{5}{2} R = 5R$$

$$\text{using } C_m = C_{v,m} + \frac{R}{1-x}$$

$$\Rightarrow \frac{5}{2} = \frac{1}{1-x}$$

$$\Rightarrow x = \frac{3}{5} = 0.60$$

2. Ans (40.00 to 40.20)



$$Q_p = \frac{P_{\text{COCl}_2}^1}{P_{\text{CO}}^1 \cdot P_{\text{Cl}_2}^1} = \frac{1/4}{1/4 \times 2/4} = 2$$

$$\Delta_r G = \Delta_r G^\circ + 2.303 RT \log Q$$

$$= 2.303 RT \log Q/K$$

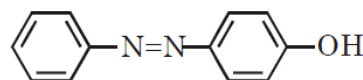
$$= 2.303 \times 8.3 \times 300 \times (-7)$$

$$= -40.141 \text{ kJ/mole}$$

$$\simeq -40.14 \text{ kJ/mole}$$

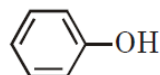
4. Ans (1.50)

Product C is



(Double bond equivalent = 9)

Product B is



(No. Of hydrogen atoms = 6)

Now

$$\frac{x}{y} = \frac{9}{6} = 1.50$$

6. Ans (825.00)



$$\Delta_r H^\circ = \sum \Delta_f H^\circ_{\text{products}} - \sum \Delta_f H^\circ_{\text{reactants}}$$

$$= [3(-75) + (-100 - 300) + 2(-300)] - [4(-100)]$$

$$= -825 \text{ kJ/mol}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C)

Let A_1 denote the event that the policy-holder will have an accident within a year of purchasing the policy, and let A denote the event that the policyholder is accident prone. Hence, the desired probability is given by
 $P(A_1) = P(A_1 | A)P(A) + P(A_1 | A^c)P(A^c)$
 $= (.4)(.3) + (.2)(.7) = .26$

2. Ans (A,B,C,D)

$(1 + 2x + 3x^2)^{10} = [1 + x(2 + 3x)]^{10}$
 $= 1 + {}^{10}C_1 x(2 + 3x) + {}^{10}C_2 x^2 (2 + 3x)^2 + \dots$
 Therefore $a_1 = {}^{10}C_1 \times 2 = 20$
 and $a_2 = {}^{10}C_2 \times 2^2 + 3 \times {}^{10}C_1 = 180 + 30 = 210$
 $x = 1 \Rightarrow a_0 + a_1 + a_2 + \dots + a_{20} = 6^{10} \quad \dots (1)$
 $x = -1 \Rightarrow a_0 - a_1 + a_2 - \dots + a_{20} = 2^{10} \quad \dots (2)$
 $(1) + (2) \Rightarrow a_0 + a_2 + \dots + a_{20} = \frac{6^{10} + 2^{10}}{2}$
 $(1) - (2) \Rightarrow a_1 + a_3 + \dots + a_{19} = \frac{6^{10} - 2^{10}}{2}$

3. Ans (A,C)

Required tangent equation is of the form $y = x + c$, $A = (3,2)$ and $r = 2\sqrt{2}$, respectively, are the centre and the radius of the given circle. Therefore, the line $y = x + c$ touches the given circle implies that

$$\frac{|3 - 2 + c|}{\sqrt{1^2 + 1^2}} = 2\sqrt{2}$$

$$\Rightarrow c + 1 = \pm 4$$

Therefore, $c = 3, -5$. Hence, the required tangents are $y = x + 3$ and $y = x - 5$

4. Ans (A,B)

All lines passes through point of intersection of $3x + 4y + 6 = 0$ and $x + y + 2 = 0$
 $4x + 4y + 8 = 0$
 $3x + 4y + 6 = 0$
 $x + 2 = 0 \Rightarrow x = -2, y = 0, P(-2, 0)$
 So greatest distance is $\frac{8 + 9 + 8}{5} = 5 \Rightarrow$ line is
 $4x + 3y + 8 = 0$

5. Ans (A,B,D)

$$P(E_1 \cap F) = P(\{4, 2\}) = \frac{1}{36}$$

$$P(E_1) = \frac{5}{36}$$

$$P(F) = \frac{1}{6}$$

6. Ans (B,D)

$$\frac{T_3}{T_2} = 7$$

$$\text{This implies } \left(\frac{n-2+1}{2} \right) \left(\frac{2^{-2x}}{2^x} \right) = 7$$

$$\Rightarrow \frac{n-1}{2} \cdot 2^{-3x} = 7 \quad \dots (1)$$

$$\text{Also } {}^nC_1 + {}^nC_2 = 36$$

$$\Rightarrow n^2 + n - 72 = 0$$

$$(n+9)(n-8) = 0$$

$$n = 8$$

Putting the value $n = 8$ in equation (1), we have

$$\frac{7}{2} \cdot 2^{-3x} = 7 \Rightarrow 2^{-3x} = 2$$

$$\text{Therefore } x = \frac{-1}{3}$$

7. Ans (B)

Total number of ways

$${}^{13+3-1}C_{3-1} = {}^{15}C_2 = 105 \text{ ways}$$

but it will also contain the ways when exactly two persons receive equal number of apples & those ways are

$$0 \quad 0 \quad 13$$

$$1 \quad 1 \quad 11$$

$$\vdots$$

$$6 \quad 6 \quad 1$$

$$\text{Number of ways} = 7 \times 3 = 21$$

$$\Rightarrow \text{Required ways} = 105 - 21 = 84 \text{ ways}$$

8. Ans (B)

$$\left(\frac{y}{x} \right)^3 + c \left(\frac{y}{x} \right)^2 + b \left(\frac{y}{x} \right) + 1 = 0$$

$$m_1, m_2, m_3 = -1 \Rightarrow m_3 = 1$$

$$m_1 m_2 + m_2 m_3 + m_3 m_1 = b$$

$$\Rightarrow -1 + (m_1 + m_2) = b \Rightarrow m_1 + m_2 = b + 1$$

$$m_1 + m_2 + m_3 = -c \Rightarrow b + c = -2$$

PART-3 : MATHEMATICS

SECTION-I (ii)

9. Ans (A)

(I) Let $S = x^2 + y^2 - 3x + 2y - 4 = 0$ and
 $L = 2x + 5y + 2 = 0$. The required circle is of the form

$$S + \lambda L = x^2 + y^2 - 3x + 2y - 4 + \lambda(2x + 5y + 2) = 0$$

$$\text{whose centre is } \left(\frac{3 - 2\lambda}{2}, \frac{-(5\lambda + 2)}{2} \right)$$

which lies on the line $x + y - 11 = 0$. That is,

$$3 - 2\lambda - 5\lambda - 2 - 22 = 0$$

$$\Rightarrow -7\lambda - 21 = 0$$

$$\Rightarrow \lambda = -3$$

Therefore, the required circle is $x^2 + y^2 - 9x - 13y - 10 = 0$.

(II) The circle described on the line joining the points $(0, 3)$ and $(-2, -1)$ as diameter is

$$x(x + 2) + (y - 3)(y + 1) = 0$$

$$\Rightarrow x^2 + y^2 + 2x - 2y - 3 = 0 \quad \dots(i)$$

The required circle is of the form $(x - 1)^2 + (y + 1)^2 + \lambda(2x + 3y + 1) = 0$

$$\Rightarrow x^2 + y^2 + 2(\lambda - 1)x + 2\left(\frac{3\lambda}{2} + 1\right)y + 2 + \lambda = 0 \dots(ii)$$

The circle given in Eq. (ii) cuts orthogonally the circle given in Eq. (i). This implies that

$$2(\lambda - 1)(1) + 2\left(\frac{3\lambda}{2} + 1\right)(-1) = 2 + \lambda - 3$$

$$\Rightarrow -\lambda - 4 = \lambda - 1$$

$$\Rightarrow \lambda = \frac{-3}{2}$$

Therefore, the required circle is

$$x^2 + y^2 - 5x - \frac{5}{2}y + 2 - \frac{3}{2} = 0$$

$$\Rightarrow 2x^2 + 2y^2 - 10x - 5y + 1 = 0$$

(III) Let $S = x^2 + y^2 - 4 = 0$ and

$$S' = x^2 + y^2 - 2x - 4y + 4 = 0.$$

Let $L = S - S' \equiv 2x + 4y - 8 = 0$. That is,

$$L \equiv x + 2y - 4 = 0$$

Now, the required circle is

$$S + \lambda L = x^2 + y^2 - 4 + \lambda(x + 2y - 4) = 0$$

$$\equiv x^2 + y^2 + 2\left(\frac{\lambda}{2}\right)x + 2\lambda y - 4 - 4\lambda = 0$$

Therefore, the centre is $\left(\frac{-\lambda}{2}, -\lambda\right)$

and the radius is $\sqrt{\frac{\lambda^2}{4} + \lambda^2 + 4 + 4\lambda}$

$S + \lambda L = 0$ touches the line $x + 2y - 5 = 0$.

This implies that

$$\frac{\left| \frac{-\lambda}{2} + 2(-\lambda) - 5 \right|}{\sqrt{1^2 + 2^2}} = \sqrt{\frac{\lambda^2}{4} + \lambda + 4 + 4\lambda}$$

$$\Rightarrow \frac{|5\lambda + 10|}{2\sqrt{5}} = \frac{1}{2}\sqrt{5\lambda^2 + 16\lambda + 16}$$

$$\Rightarrow 5(\lambda + 2)^2 = 5\lambda^2 + 16\lambda + 16$$

$$\Rightarrow 4\lambda + 4 = 0$$

$$\Rightarrow \lambda = -1$$

$\lambda = -1$ gives the circle $x^2 + y^2 - x - 2y = 0$

(IV) $ax + by + c = 0$ and $2b = a + c$

$$a - 2b + c = 0$$

Point of concurrency of family = $(1, -2)$

Which is centre of circle

Let equation of circle is

$$x^2 + y^2 - 2x + 4y + c = 0 \text{ which is orthogonal}$$

$$\text{to } x^2 + y^2 - 4x - 4y - 1 = 0$$

$$\text{So } 2(-1)(-2) + 2(2)(-2) = c - 1$$

$$c = -3$$

$$\text{so circle is } x^2 + y^2 - 2x + 4y - 3 = 0$$

10. **Ans (D)**

(I) Let $S = x^2 + y^2 - 3x + 2y - 4 = 0$ and
 $L = 2x + 5y + 2 = 0$. The required circle is of the
 form
 $S + \lambda L = x^2 + y^2 - 3x + 2y - 4 + \lambda(2x + 5y + 2) = 0$
 whose centre is $\left(\frac{3-2\lambda}{2}, \frac{-(5\lambda+2)}{2}\right)$
 which lies on the line $x + y - 11 = 0$. That is, $3 - 2\lambda$
 $- 5\lambda - 2 - 22 = 0$
 $\Rightarrow -7\lambda - 21 = 0$
 $\Rightarrow \lambda = -3$

Therefore, the required circle is

$$x^2 + y^2 - 9x - 13y - 10 = 0.$$

(II) The circle described on the line joining the
 points (0, 3) and (-2, -1) as diameter is

$$x(x+2) + (y-3)(y+1) = 0$$

$$\Rightarrow x^2 + y^2 + 2x - 2y - 3 = 0 \quad \dots(i)$$

The required circle is of the form $(x-1)^2 + (y+1)^2 + \lambda(2x+3y+1) = 0$
 $\Rightarrow x^2 + y^2 + 2(\lambda-1)x + 2\left(\frac{3\lambda}{2} + 1\right)y + 2 + \lambda$
 $= 0 \dots(ii)$

The circle given in Eq. (ii) cuts orthogonally
 the circle given in Eq. (i). This implies that

$$2(\lambda-1)(1) + 2\left(\frac{3\lambda}{2} + 1\right)(-1) = 2 + \lambda - 3$$

$$\Rightarrow -\lambda - 4 = \lambda - 1$$

$$\Rightarrow \lambda = \frac{-3}{2}$$

Therefore, the required circle is

$$x^2 + y^2 - 5x - \frac{5}{2}y + 2 - \frac{3}{2} = 0$$

$$\Rightarrow 2x^2 + 2y^2 - 10x - 5y + 1 = 0$$

(III) Let $S = x^2 + y^2 - 4 = 0$ and

$$S' = x^2 + y^2 - 2x - 4y + 4 = 0.$$

Let $L = S - S' \equiv 2x + 4y - 8 = 0$. That is,

$$L \equiv x + 2y - 4 = 0$$

Now, the required circle is

$$S + \lambda L = x^2 + y^2 - 4 + \lambda(x + 2y - 4) = 0$$

$$\equiv x^2 + y^2 + 2\left(\frac{\lambda}{2}\right)x + 2\lambda y - 4 - 4\lambda = 0$$

Therefore, the centre is $\left(\frac{-\lambda}{2}, -\lambda\right)$

and the radius is $\sqrt{\frac{\lambda^2}{4} + \lambda^2 + 4 + 4\lambda}$

$S + \lambda L = 0$ touches the line $x + 2y - 5 = 0$. This
 implies that

$$\frac{\left|\frac{-\lambda}{2} + 2(-\lambda) - 5\right|}{\sqrt{1^2 + 2^2}} = \sqrt{\frac{\lambda^2}{4} + \lambda^2 + 4 + 4\lambda}$$

$$\Rightarrow \frac{|5\lambda + 10|}{2\sqrt{5}} = \frac{1}{2}\sqrt{5\lambda^2 + 16\lambda + 16}$$

$$\Rightarrow 5(\lambda + 2)^2 = 5\lambda^2 + 16\lambda + 16$$

$$\Rightarrow 4\lambda + 4 = 0$$

$$\Rightarrow \lambda = -1$$

$\lambda = -1$ gives the circle $x^2 + y^2 - x - 2y = 0$

(IV) $ax + by + c = 0$ and $2b = a + c$

$$a - 2b + c = 0$$

Point of concurrency of family = (1, -2)

Which is centre of circle

Let equation of circle is

$x^2 + y^2 - 2x + 4y + c = 0$ which is orthogonal
 to $x^2 + y^2 - 4x - 4y - 1 = 0$

$$\text{So } 2(-1)(-2) + 2(2)(-2) = c - 1$$

$$c = -3$$

so circle is $x^2 + y^2 - 2x + 4y - 3 = 0$

11. **Ans (A)**

(I) AL CHEMIST

$\Rightarrow$ No of arrangements is $\underline{3}$

(II) First and last position can be filled in

$\left({}^3C_2 \cdot 2!\right)$ ways and Rest in 7!

$$\Rightarrow \text{No of ways} = {}^3C_2 \cdot 7! \cdot 2!$$

(III) A, E, I are filled in first four position

$$\Rightarrow \text{No of ways} = {}^4C_3 \cdot \underline{3} \cdot \underline{6} = \underline{4} \cdot \underline{6}$$

(IV) There are five odd positive

$$\Rightarrow \left({}^5C_3 \cdot \underline{3}\right) \cdot \underline{6}$$

12. **Ans (C)**

(I) AL CHEMIST

⇒ No of arrangements is $\underline{3}$

(II) First and last position can be filled in $({}^3C_2 \cdot 2!)$ ways and Rest in 7!

⇒ No of ways = ${}^3C_2 \cdot 7! \cdot 2!$

(III) A, E, I are filled in first four position

⇒ No of ways = ${}^4C_3 \cdot \underline{3} \cdot \underline{6} = \underline{4} \cdot \underline{6}$

(IV) There are five odd positive

⇒ $({}^5C_3 \cdot \underline{3}) \cdot \underline{6}$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (0.56)**

Line AB is $y - 2 = 1(x - 1)$

⇒ $x - y + 1 = 0$ (1)

and $2x + 3y = 5$ (2)

on solving (1) & (2)

⇒ $5x = 2$

⇒ $h = \frac{2}{5}$ and $k = \frac{7}{5}$

2. **Ans (0.25)**

(1) (3 consecutive heads)

HHHTT, THHHT, TTHHH, HTHHH,
HHHHTH = $\frac{5}{32}$

(2) (4 consecutive heads)

HHHHT, THHHH = $\frac{2}{32}$

(3) (5 consecutive heads)

HHHHH = $\frac{1}{32}$

Ans = $\frac{1}{4}$

3. **Ans (1.10)**

We have

$$11! \left[\frac{1}{1!10!} + \frac{1}{3!8!} + \frac{1}{5!6!} + \dots + \frac{1}{11!1!} \right]$$

$$= \frac{11}{1!10!} + \frac{11}{3!8!} + \frac{11}{5!6!} + \dots + \frac{11}{11!}$$

$$= {}^{11}C_1 + {}^{11}C_3 + {}^{11}C_5 + {}^{11}C_7 + {}^{11}C_9 + {}^{11}C_{11}$$

= sum of the even coefficient in the expansion of $(1+x)^{11} = 2^{10}$

Therefore $K = 10, n = 11$

4. **Ans (0.17)**

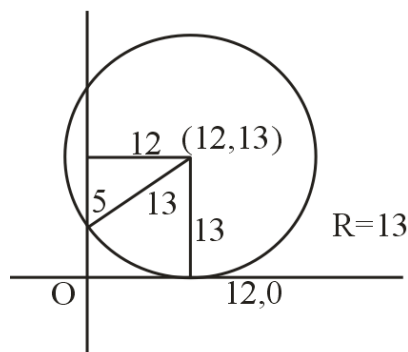
In this case

$n = 10; p = 1/2; q = 1/2$

$P(X \geq 7) = P(X=7) + P(X=8) + P(X=9) + P(X=10)$

$$= ({}^{10}C_3 + {}^{10}C_2 + {}^{10}C_1 + 1) \frac{1}{2^{10}} = \frac{176}{2^{10}} = \frac{11}{64}$$

5. **Ans (13.00)**



6. **Ans (13.60)**

Each of the 30 integers can fall into following set, having 10 elements in each.

To; $\{3, 6, 9, 12, 15, 18, 21, 24, 27, 30\} \rightarrow 3k$

To; $\{1, 4, 7, 10, 13, 16, 19, 22, 25, 28\} \rightarrow 3k + 1$

To; $\{2, 5, 8, 11, 14, 17, 20, 23, 26, 29\} \rightarrow 3k + 2$

Consider a general subset $\{x, y, z\}$ of S.

So that $x + y + z = 3$

Because multiple of 3 can be formed out of 3 reminders in exactly four different ways. $0 + 0 + 0, 1 + 1 + 1, 2 + 2 + 2, 0 + 1 + 2.$

⇒ Either x, y, z selected from same set (T_0, T_1, T_2) or

Exactly one of them lie in each of the these sets (T_0, T_1, T_2)

$$\Rightarrow P = 3 \times {}^{10}C_3 + ({}^{10}C_1 \cdot {}^{10}C_1 \cdot {}^{10}C_1) = 1360$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

PART TEST

07-01-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I & I(N)] : (SCORE-I)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	B	C	D	B	B	B	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	D	D	C	D	C	D	B	C	D
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5.00	9.00	2.00	1.10	1.57	0.75	2.00	81.00	8.00	4.20

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	B	A	D	D	B	C	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	D	D	B	A	C	A	D	A	C
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5.00	1.00	1.00	3.00	5.00	2.00	4.00	4.00	5.00	3.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	B	A	C	C	A	C	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	A	B	A	C	B	B	D	B	B
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.00	0.09	2.50	0.25	504.00	16.00	6.00	18.25	2.00	0.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I
1. Ans (A)

$$\frac{\lambda}{2\pi\epsilon_0 x} = \frac{2q\ell}{2\pi\epsilon_0 x^3}$$

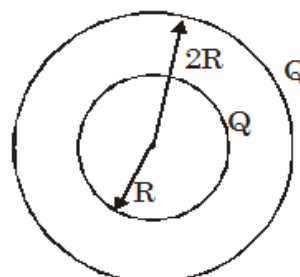
$$x^2 = \frac{2q\ell}{\lambda}; x = 2$$

2. Ans (B)

Place another identical hemisphere on given one, so that flat faces of two coincide. The total flux linked with cross-section of bottom half (lower hemisphere) is $\frac{Q}{2\epsilon_0}$ as charge enclosed by given hemisphere is zero. So flux linked with flat face of hemisphere is $\frac{-Q}{2\epsilon_0}$

3. Ans (B)

All the spheres are at same potential So, $V_1 = V_2 = V_3$ and $E_1 : E_2 : E_3 = \frac{1}{R_1} : \frac{1}{R_2} : \frac{1}{R_3}$

4. Ans (C)


$$U = \frac{kQ^2}{2R} + \frac{kQ^2}{2 \times 2R} + \frac{kQ}{2R} \times Q$$

$$\frac{kQ^2}{2R} + \frac{kQ^2}{4R} + \frac{kQ^2}{2R}$$

$$\frac{kQ^2}{R} + \frac{kQ^2}{4R} = \frac{5}{4} \frac{kQ^2}{R}$$

5. **Ans (D)**

$$V_B - V_A = - \int E_x dx = - (\text{area under } E_x - x \text{ curve})$$

$$V_B - 10 = - \frac{1}{2} \times 2 \times (-20) = 20$$

$$V_B = 30V$$

7. **Ans (B)**

$$E_1 = \frac{KP_1}{r^3}$$

$$F = P_2 \left| \frac{dE_1}{dr} \right| = \frac{3KP_1P_2}{r^4}$$

8. **Ans (B)**

$$g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$$

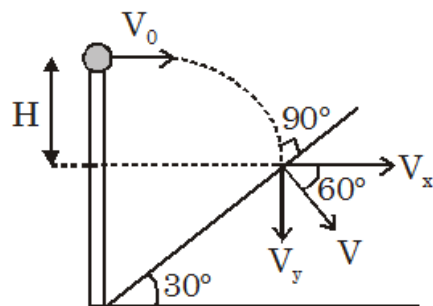
9. **Ans (D)**

$$\text{Radius of curvature} = \frac{v^2}{a_{\perp}}$$

10. **Ans (C)**

use the definition of kinematics variable

11. **Ans (A)**



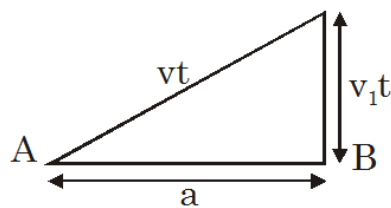
$$v_y = \sqrt{2gH}$$

$$\frac{v_y}{v_x} = \frac{\sqrt{2gH}}{v_0} = \tan 60^\circ = \sqrt{3}$$

$$v_0 = \sqrt{\frac{2gH}{3}}$$

12. **Ans (D)**

$$a^2 + v_1^2 t^2 = v^2 t^2$$



$$t = \frac{a}{\sqrt{v^2 - v_1^2}}$$

13. **Ans (D)**

$$kx = \frac{N}{\sqrt{2}} = mg, x = \frac{mg}{k}$$

14. **Ans (C)**

Acceleration of block = -3 m/s^2

Acceleration of plank = -7 m/s^2

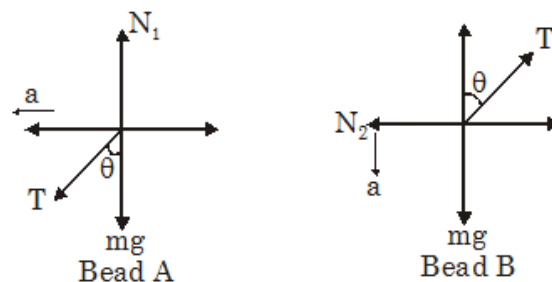
After 3 sec. the plank will come to rest. In 3 sec. the block will move $\frac{1}{2} \times 4 \times 3 \times 3 = 18$ m relative to the plank.

Now let the block will cover remaining 18 m relative to plank in next t seconds.

$$12t - \frac{1}{2} \times 3t^2 = 18$$

So t = 2 sec, therefore total time is 5 sec

15. **Ans (D)**



$$T \sin \theta = ma \quad \text{--- (i)}$$

$$mg - T \cos \theta = ma$$

$$T = \frac{mg}{\sin \theta + \cos \theta} = \frac{mg}{\sqrt{2}} \quad (\theta = 45^\circ)$$

16. **Ans (C)**

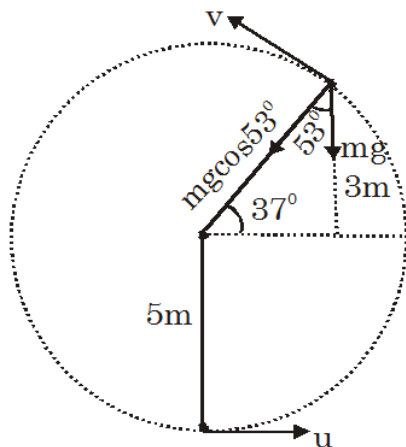
Work done = area under p - t graph as t increases, area increases

17. **Ans (D)**

$$w_F + w_{\text{gravity}} = 0$$

$$w_F - mgH = 0, w_F = mgH$$

18. Ans (B)



$$mg \cos 53^\circ = \frac{mv^2}{5}$$

$$v^2 = 5g \cos 53^\circ$$

$$= 5g \times \frac{3}{5} = 3g$$

by energy conservation

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mg \times 8$$

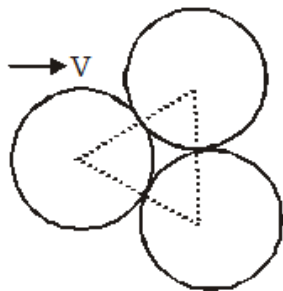
$$u^2 = v^2 + 16g$$

$$u^2 = 3g + 16g = 19g$$

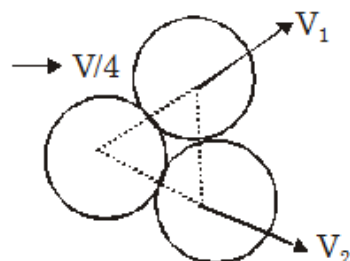
$$u = \sqrt{190} \text{ ms.}$$

19. Ans (C)

Before collision



After collision



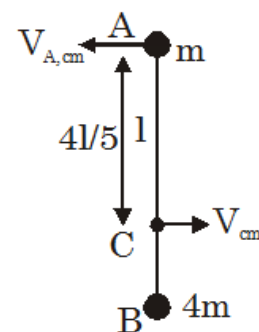
$$2mv_1 \cos 30^\circ + m\frac{V}{4} = mv \text{ ----- (i)}$$

$$e = \frac{v_1 - \frac{V}{4} \cos 30^\circ}{V \cos 30^\circ} \text{ ----- (ii)}$$

20. Ans (D)

$$J = 5m \times V_{cm}$$

In the COM frame, particles will be in circular motion.



$$T = \frac{m(V_{A,cm})^2}{(4l/5)} = \frac{m\left(\frac{J}{5m}\right)^2}{(4l/5)} = \frac{J^2}{20ml}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (5.00)

$$\phi = \frac{q}{24\epsilon_0} - \frac{q}{\epsilon_0 4\pi} \frac{2\pi \left(1 - \frac{1}{\sqrt{2}}\right)}{4}$$

$$\frac{q}{24\epsilon_0} - \frac{q}{8\epsilon_0} + \frac{q}{8\sqrt{2}\epsilon_0} = \frac{q}{8\sqrt{2}\epsilon_0} - \frac{q}{12\epsilon_0}$$

$$= 5 \frac{Nm^2}{c}$$

2. Ans (9.00)

$$\frac{1}{2}mv^2 + \frac{kQq}{2R} = \frac{kQq}{R}$$

$$\frac{1}{2}mv^2 = \frac{kQq}{2R}$$

$$V = \sqrt{\frac{kQq}{mR}} = \sqrt{\frac{9 \times 10^9 \times 6 \times 10^{-6} \times 3 \times 10^{-6}}{2 \times 10^{-3} \times 1}}$$

$$= \sqrt{9 \times 9} = 9 \text{ m/s.}$$

3. Ans (2.00)

We have relation $M \frac{dv}{dt} = F_{\text{ext}} + V_{\text{rel}} \frac{dM}{dt}$

In the question, we have the force by the gravity as the external force. As the rocket is to lift from the ground the minimum acceleration of the rocket to do so is zero. beyond which with a slight increase it will zoom into the sky.

Therefore, $\frac{dv}{dt} = 0$

Putting these into the equation, we obtain

$$0 = Mg + v_{\text{rel}} - \left(\frac{dM}{dt} \right)$$

($\frac{dM}{dt}$ is rate of decrease of mass hence negative. Mg is in the direction of the relative velocity of the fuel with respect to the rocket)

Hence, $\frac{dM}{dt} = \frac{Mg}{V} = \frac{400 \times 10}{2 \times 1(10^3)} = 2 \text{ kg/sec}$

4. Ans (1.10)

$$\frac{1}{2} \mu u^2 = mgR$$

$u = \sqrt{2gR} \rightarrow$ Velocity of particle at the lowest point

$\mu u = 2mv \Rightarrow v$ velocity of both at highest point

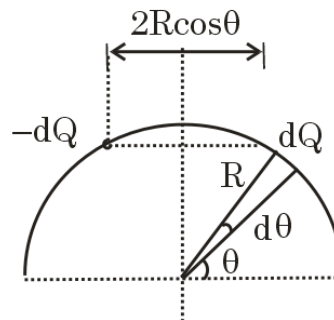
$$v = u/2$$

$$\frac{1}{2} \mu u^2 = mgh + \frac{1}{2} \times 2m \times v^2$$

on solving $h = \frac{R}{2}$

$$= \frac{2.2}{2} = 1.1 \text{ m}$$

5. Ans (1.57)



$$dp = dQ \cdot 2R \cos \theta$$

$$= \lambda_0 \cos \theta R d\theta \cdot 2R \cos \theta$$

$$= 2\lambda_0 R^2 \int_0^{\pi/2} \cos^2 \theta d\theta$$

$$\cos 2\theta = 2\cos^2 \theta - 1$$

$$= 2\lambda_0 R^2 \int_0^{\pi/2} \frac{(1 + \cos 2\theta)}{2} = \lambda_0 R^2 \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\pi/2}$$

$$= \frac{\lambda_0 R^2 \pi}{2}$$

6. Ans (0.75)

$f_{s,\text{max}} = 20 \text{ N}$. a_{max} of 20 kg block = 1 m/s^2

$\therefore$ both blocks move together till 6 sec.

$$\frac{dv}{dt} = \frac{5t}{30} \Rightarrow \int dv = \frac{1}{6} t^2$$

$$V = \frac{3^2}{12} = \frac{9}{12} = 0.75 \text{ m/s}$$

7. Ans (2.00)

Let the length of rod is ℓ and co- ordinate of B is (x, y)

$$\vec{v}_B = v_x \hat{i} + v_y \hat{j} = \sqrt{3} \hat{i} + v_y \hat{j}$$

$$x^2 + y^2 = \ell^2$$

$$\Rightarrow 2xv_x + 2yv_y = 0 \Rightarrow \sqrt{3} + \frac{y}{x} v_y = 0$$

$$\Rightarrow \sqrt{3} + \tan 60^\circ v_y = 0 \quad v_y = -1 \text{ m/s}$$

$$\vec{v}_B = \sqrt{3} \hat{i} - 1 \hat{j}$$

$$|\hat{v}_B| = \sqrt{3+1} = 2 \text{ m/s}$$

8. Ans (81.00)

$$u = \frac{2kQq}{\ell} = \frac{2kQq}{R/\sin 37^\circ} = \frac{2kQq \sin 37^\circ}{R}$$

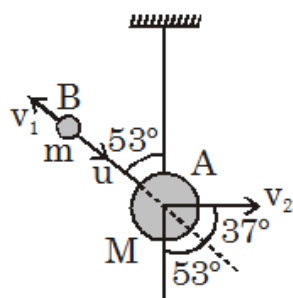
9. **Ans (8.00)**

Conserving angular momentum of the package about the centre of the planet and conserving mechanical energy.

10. **Ans (4.20)**

Using conservation of momentum along horizontal direction

$$mu \sin 53^\circ = -mv_1 \sin 53^\circ + Mv_2$$



$$\frac{4}{5}u = -v_1 \frac{4}{5} + 4v_2$$

$$5v_2 = 2u$$

$$\frac{v_2 \cos 37^\circ + v_1}{u} = e = 1 \quad \text{-----(i)}$$

$$0.8v_2 + v_1 = u \quad \text{-----(ii)}$$

Solving (i) and (ii), we get

$$5.8v_2 = 2u$$

$$v_2 = \frac{2u}{5.8} = 2\text{m/s}$$

$$\therefore v_1 = 5v_2 - u = 5 \times 2 - 5.8 = 4.2\text{m/s}$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (B)**

B - N bonds are polar in $B_3N_3H_6$

2. **Ans (D)**

Xe > Kr > Ar > Ne > He (Ease of liquification)

3. **Ans (B)**

(A) Pyrophosphorous acid ($H_4P_2O_5$)

(B) Hypophosphoric acid ($H_4P_2O_6$)

(C) Pyrophosphoric acid ($H_4P_2O_7$)

(D) cyclotrimeta phosphoric acid (HPO_3)₃

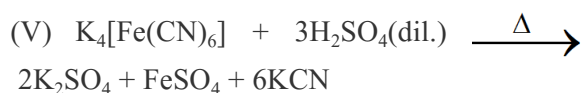
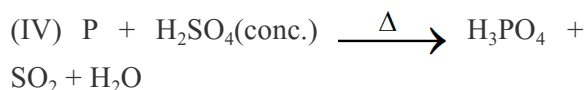
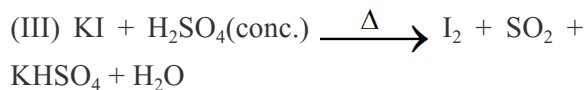
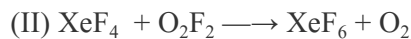
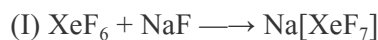
5. **Ans (D)**

Graphite have sp^2 while diamond has sp^3 hybridised 'C' atom.

6. **Ans (D)**

$XeOF_4$ and XeO_2F_2 are obtained by partial hydrolysis of XeF_6 .

7. **Ans (B)**

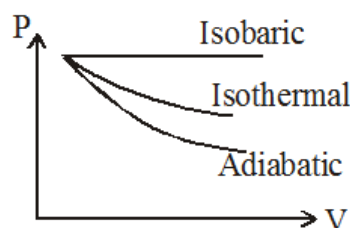


8. **Ans (C)**

Work done by the gas in the cyclic process = Area bounded (ABCA) = $-5 P_1 V_1$

Therefore, $q = -W = 5 P_1 V_1$ (cyclic process, so $E = 0$).

9. **Ans (B)**



Work is area under P- V curve.

10. **Ans (C)**

Since work done on the system, is maximum in single stage compression hence all other adiabatic points will fall between 1 and 2.

11. **Ans (B)**

$$\Delta S_{\text{system}} = nR \ln \frac{V_2}{V_1} = 2 \times R \times \ln 2 = 11.52 \text{ J/K}$$

$$\Delta S_{\text{surrounding}} = - \frac{3.41 \times 1000}{310} = -11 \text{ J/K}$$

$$\Delta S_{\text{total}} = +11.52 - 11 = +0.52 \text{ J/K}$$

13. **Ans (D)**

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -RT \ln K_{\text{eq}} = 0 \text{ [K}_{\text{eq}} = 1]$$

$$\Rightarrow T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{200 \times 10^3}{40 \times 10^3} = 5\text{K}$$

14. **Ans (B)**

$$\Delta G^\circ = \Delta H^\circ - \Delta TS^\circ$$

$$\Delta H_r^\circ = (\sum \Delta_f H^\circ)_p - (\sum \Delta_f H^\circ)_R$$

$$= 2 \times (+40.407) - (+70)$$

$$= +10.814 \text{ kJ}$$

$$\Delta G^\circ = +10.814 \text{ kJ} - 250 \times 10 \text{ J/K}$$

$$= +10.814 \text{ kJ} - 2500 \text{ J} = 8314 \text{ J}$$

$$\Delta G^\circ = -RT \ln K$$

$$8314 = -8.314 \times 250 \ln K$$

$$\ln K = -4.$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (5.00)**

a, c, d, e, h

5. **Ans (5.00)**

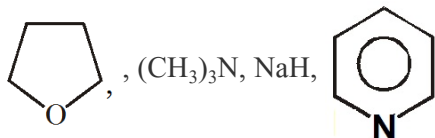
C > Si > Ge > Pb > Sn (IE_1)

6. **Ans (2.00)**

cyclic silicate, single chain silicate

7. **Ans (4.00)**

symmetrical cleavage $\rightarrow$ Large lewis base $\rightarrow$



unsymmetrical cleavage $\rightarrow$ small lewis base

$\rightarrow$ NH_3 , CH_3NH_2 , $(\text{CH}_3)_2\text{NH}$,

8. **Ans (4.00)**

[(ii), (iii), (vii), (viii)].

9. **Ans (5.00)**

$$W = -P \Delta V$$

$$W = -nR \Delta T$$

$$\text{or } n = \frac{4.157 \times 10^3}{8.314 \times 100} = 5$$

10. **Ans (3.00)**

$P_{(\text{white})}$, Cu(s) , $\text{H}^+(\text{aq})$ have ΔH_f° at 298 K = 0.

$P_{(\text{osr})}$, Cu(s) , $\text{H}^+(\text{aq})$ 298 K ij $\Delta H_f^\circ = 0$ j[krs
gSaA

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A)**

$$P\left(\frac{3k}{2}, 1\right)$$

$$x + y = k, x + y = 2k$$

$$\text{or } \left(\frac{3k}{2} + 1 - k\right) \left(\frac{3k}{2} + 1 - 2k\right) < 0$$

$$\text{or } \left(\frac{k}{2} + 1\right) \left(1 - \frac{k}{2}\right) < 0$$

$$\text{or } \left(\frac{k}{2} + 1\right) \left(\frac{k}{2} - 1\right) > 0$$

$$\begin{array}{ccccccc} & + & & - & & + & \\ & & -2 & & +2 & & \end{array}$$

$$k < -2 \text{ or } k > 2$$

$$\text{so } k = 3$$

2. **Ans (B)**

We have to distribute the three groups of 6 members each to 3 persons and this can be

$$\text{done in } \frac{18!}{(6!)^3 3!}$$

3. **Ans (B)**

Combined equation of OA & OB is

$$x^2 + y^2 = a^2 \left(\frac{x+y}{3}\right)^2$$

$$\Rightarrow 9x^2 + 9y^2 = a^2(x^2 + y^2 + 2xy)$$

$$\therefore \text{OA} \perp \text{OB}$$

$$\Rightarrow \text{coeff. of } x^2 + \text{coeff. of } y^2 = 0$$

$$\Rightarrow a^2 = 9$$

4. **Ans (A)**

$$1 - \frac{{}^6C_1 \cdot 5^3}{6^4} = \frac{91}{216}$$

5. **Ans (C)**

The given equation can be written as

$$(2x + 3y - 5) + \tan \phi (3x - 5y + 2) = 0$$

The lines $2x + 3y - 5 = 0$ and

$$3x - 5y + 2 = 0 \text{ intersects at } (1, 1)$$

The reflection of (1,1) with respect to the line

$$x + y = \sqrt{2} \text{ is } (\sqrt{2} - 1, \sqrt{2} - 1)$$

6. **Ans (C)**

$$P(A) = \frac{1}{3}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{3}{4}$$

$$\begin{aligned} P(B) &= \frac{3}{4} - \frac{1}{3} + P(A \cap B) \\ &= \frac{5}{12} + \left[0, \frac{1}{3}\right] \\ &= \left[\frac{5}{12}, \frac{3}{4}\right] \end{aligned}$$

7. **Ans (A)**

Let the circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

(i)

passing through (a, b)

$$\text{So, } a^2 + b^2 + 2ga + 2fb + c = 0 \quad \dots(ii)$$

Circle (i) is intersecting the circle $x^2 + y^2 = 4$ orthogonally

$$\text{So, } 2g_1g_2 + 2f_1f_2 = C_1 + C_2$$

$$0 + 0 = C - 4$$

$$C = 4$$

from (ii)

$$a^2 + b^2 - 2a(-g) - 2f(-b) + 4 = 0$$

Locus of centre is

$$a^2 + b^2 - 2ax - 2fy + 4 = 0$$

$$2ax + 2fy - (a^2 + b^2 + 4) = 0$$

8. **Ans (C)**

Let B_1 = put on a Black sock.

& B_2 = put on a Brown sock.

$$P(B_1) = \frac{6}{10} = \frac{3}{5}$$

$$P(B_2|B_1) = \frac{4}{9}$$

$$\begin{aligned} \therefore P(B_1 \cap B_2) &= P(B_1)P(B_2|B_1) \\ &= \frac{3}{5} \cdot \frac{4}{9} = \frac{4}{15} \end{aligned}$$

9. **Ans (B)**

$$T_{r+1} = {}^nC_r (2x)^{n-r} (5y)^r$$

$$T_r = {}^nC_{r-1} (2x)^{n-r+1} (5y)^{r-1}$$

$$\frac{T_{r+1}}{T_r} = \frac{(n-r+1)}{r} = \left(\frac{34-r+1}{r}\right)$$

$$\left(\frac{10}{6}\right) > 1$$

$$\Rightarrow \frac{35-r}{r} > \frac{3}{5}$$

$$3r < 175 - 5r \quad 8r < 175$$

$$r < \left(\frac{175}{8}\right) \quad r = 0, 1, \dots, 21$$

$\Rightarrow T_{22}$ is largest.

10. **Ans (A)**

Let $A = \{a_1, a_2, a_3, a_4, a_5, a_6, a_7\}$

and four sets A_1, A_2, A_3, A_4

If A_1, A_2, A_3 and A_4 are disjoint sets then each and every element of set A have five choices.

a_1 will be in either in set A_1 or set A_2 or set A_3 or set A_4 or in none set similarly a_2 have 5 choices Each and every element of set A will have 5 choices.

Hence total ways = 5^7

11. **Ans (B)**

General term of the given expansion will be

$$\frac{5!}{n_1!n_2!n_3!} \times x^{2(n_1-n_2)} \cdot 3^{n_3}$$

we require coefficient of x^6 so

$$2(n_1 - n_2) = 6 \text{ and } n_1 + n_2 + n_3 = 5$$

$$n_1 - n_2 = 3 \text{ and } n_1 + n_2 + n_3 = 5$$

following 2 solutions are possible

$$n_1 = 3, n_2 = 0, n_3 = 2 \text{ and } n_1 = 4, n_2 = 1, n_3 = 0$$

So the coefficient of x^6 will be

$$\frac{5!}{3!2!} \times 3^2 + \frac{5!}{4!1!} \times 4^1 = 90 + 20 = 110$$

12. **Ans (A)**

Required probability

$$= 1 - \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right)$$

$$\Rightarrow \frac{3}{5}$$

13. **Ans (B)**

$$\text{As } {}^{10}C_0 = {}^{11}C_0 \text{ \& } {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

$$\text{so } \underbrace{{}^{11}C_0 + {}^{11}C_1}_{{}^{12}C_1} + {}^{12}C_2 + \dots + {}^{100}C_{90}$$

$${}^{12}C_1 + {}^{12}C_2 + {}^{13}C_3 + \dots + {}^{100}C_{90}$$

$${}^{13}C_2 + {}^{13}C_3 + \dots + {}^{100}C_{90} = {}^{101}C_{90}$$

14. **Ans (A)**

Let the line is $y = mx + c$

$$\text{OR } mx - y + C = 0$$

Here

$$p_1 = \frac{3m+c}{\sqrt{1+m^2}}, p_2 = \frac{c-3}{\sqrt{1+m^2}}, p_3 = \frac{2m-2+c}{\sqrt{1+m^2}}$$

$$\text{Given } p_1 + p_2 + p_3 = 0$$

$$5m - 5 + 3c = 0$$

$$5 = 5m + 3c$$

$$\frac{5}{3} = \frac{5}{3}m + c$$

Copare with $y = mx + c$

$$\text{Line passing through } \left(\frac{5}{3}, \frac{5}{3}\right)$$

$$a = \frac{5}{3}, b = \frac{5}{3} \text{ so } a - b = 0$$

15. **Ans (C)**

P(at least two among A,B and C)

$$= 0.8 + 0.1 - 0.6 = 0.3$$

16. **Ans (B)**

For only one point common given line $x - y + = 0$ must be the tangent of the circle

$$x^2 + y^2 + 2x - 1 = 0$$

using $p_{\perp} = r \Rightarrow$ perpendicular from the center $(-1,0)$ on the given line should be equal to the radius $\sqrt{2}$ of the given circle

$$\text{Here } \left| \frac{-1+\lambda}{\sqrt{2}} \right| = \sqrt{2}$$

$$|\lambda - 1| = 2$$

$$\lambda - 1 = \pm 2$$

$$\lambda = 3, -1$$

17. **Ans (B)**

Let the line is $y = mx + c$

$$\frac{y - mx}{c} = 1$$

$$3x^2 - y^2 - 2x \left(\frac{y - mx}{c} \right) + 4y \frac{(y - mx)}{c} = 0$$

$$x^2 \text{ coeff} + y^2 \text{ coeff} = 0$$

$$3 - 1 + \frac{2m}{c} + \frac{4}{c} = 0$$

$$2c + 2m + 4 = 0$$

$$m + c = -2$$

$$m = -2 - c$$

$$y = (-2 - c)x + c$$

$$(y + 2x) + c(x - 1) = 0$$

$$L_1 + \lambda L_2 = 0 \Rightarrow x = 1 \text{ and } y + 2x = 0$$

$$y = -2$$

$$(1, -2)$$

18. **Ans (D)**

Let $P(A) = P(B) = P$

$$P(B/A) = \frac{P(B \cap A)}{P(A)}$$

$$P(B \cap A) = P(B/A) \cdot P(A)$$

$$P(B \cap A) = (.8)P$$

$$P(A \cup B) = P + P - P(A \cap B) = (1.2)P$$

$$.6 = 1.2P$$

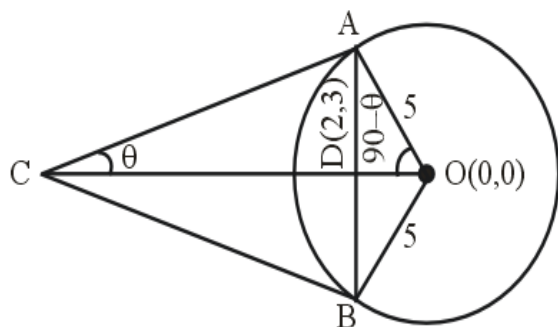
$$P = \frac{1}{2} = P(A) = P(B)$$

$$\text{Now } P[(A \cap \bar{B}) \cup (\bar{A} \cap B)]$$

$$= P(A) + P(B) - 2P$$

$$= 1 - 2 \cdot (.8) \cdot \frac{1}{2} = .2 = \frac{1}{5}$$

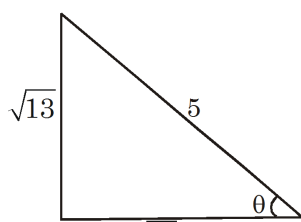
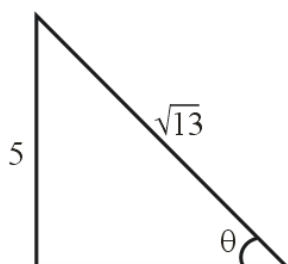
19. Ans (B)



Let D is the mid point of AB $D(2,3)$

$$OD = \sqrt{13}$$

$$\text{In } \triangle AOD, \cos(90^\circ - \theta) = \frac{\sqrt{13}}{5}$$



$$\sin \theta = \frac{\sqrt{13}}{5}$$

$$\tan \theta = \frac{\sqrt{13}}{\sqrt{12}}$$

Now In $\triangle CAO$,

$$\tan \theta = \frac{5}{CA}$$

$$\frac{5}{CA} = \frac{\sqrt{13}}{\sqrt{12}} \Rightarrow \frac{5 \times \sqrt{12}}{\sqrt{13}} = CA$$

Now Area of $OACB = CA \times 5$

$$= \frac{5\sqrt{12} \times 5}{\sqrt{13}}$$

$$= 50\sqrt{\frac{3}{13}} \text{ Sq. unit}$$

$$= 141$$

20. Ans (B)

The required no of ways

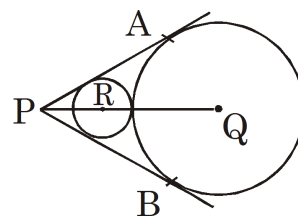
$$\begin{aligned} &= {}^{10}C_3 + {}^{10}C_4 + {}^{10}C_5 + \dots + {}^{10}C_6 + {}^{10}C_7 \\ &= 2^{10} - ({}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}) \\ &= 2^{10} - 2(1 + 10 + 45) \\ &= 912 \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.00)

Let Q be the centre of the given circle and R be the centre of the smallest circle touching the given circle externally. Then



$$AQ = 3 + 2\sqrt{2}$$

$$PQ = 3\sqrt{2} + 4$$

Let radius of smaller circle is r

$$3\sqrt{2} + 4 = QR + RP$$

$$3\sqrt{2} + 4 = 3 + 2\sqrt{2} + r + r\sqrt{2}$$

$$r = 1$$

$$\angle APQ = \theta$$

$$\sin \theta = \frac{AQ}{PQ} = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}$$

$$\angle APQ = \frac{\pi}{4}$$

$$\text{Hence } RP = r\sqrt{2}$$

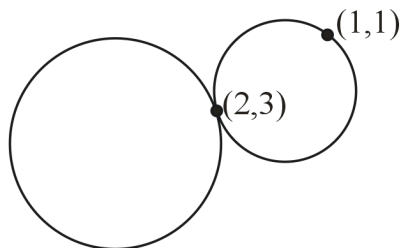
2. Ans (0.09)

$$P(\text{all tails or all heads in one trial}) = \frac{1}{4}$$

$\therefore$ P(this happens second time in 3rd trial)

$$= {}^2C_1 \times \frac{1}{4} \times \frac{3}{4} \times \frac{1}{4} = \frac{3}{32}$$

3. Ans (2.50)



Equation of the required circle is

$$(x^2 + y^2 + 4x - 6y - 3) + \lambda[(x - 2)^2 + (y - 3)^2] = 0$$

put $x = 1, y = 1$

$$\lambda = \frac{3}{5}$$

The required circle is

$$(x^2 + y^2 + 4x - 6y - 3) + \frac{3}{5}[(x - 2)^2 + (y - 3)^2] = 0$$

$$8x^2 + 8y^2 + 8x - 48y + 24 = 0$$

$$x^2 + y^2 + x - 6y + 3 = 0$$

$$r = \sqrt{\frac{1}{4} + 9 - 3} = \frac{5}{2}$$

4. Ans (0.25)

$$T_{5+1} = {}^{10}C_5 (x^2)^5 \left(\frac{1}{2x^2} \sin^{-1} 2x \right)^5$$

$$= \frac{7}{216} \left(\frac{\pi}{2} \right)^5$$

$$\Rightarrow {}^{10}C_5 \times 10 \cdot \frac{1}{2^5 x^{10}} (\sin^{-1} 2x)^5$$

$$= \frac{7}{216} \cdot \frac{\pi^5}{2^5}$$

$$\Rightarrow (\sin^{-1} 2x)^5 = \frac{7}{216} \cdot \pi^5 \cdot \frac{1}{252}$$

$$\Rightarrow (\sin^{-1} 2x)^5 = \frac{\pi^5}{6^5}$$

$$\Rightarrow \sin^{-1} 2x = \frac{\pi}{6}$$

$$\Rightarrow 2x = \sin \frac{\pi}{6} \Rightarrow x = \frac{1}{4}$$

5. Ans (504.00)

All the arrangement of the word "GENIUS" is
 $6! = 720$

Number of arrangements which either start with G or end with S is

$$= 5! + 5! - 4! = 216$$

Total number of arrangements which neither start with G nor end with S is

$$= 720 - 216 = 504$$

6. Ans (16.00)

$$\sum_{r=0}^8 {}^8C_r {}^{17}C_{16-r}$$

$${}^8C_0 {}^{17}C_{16} + {}^8C_1 {}^{17}C_{15} + \dots + {}^8C_8 {}^{17}C_8$$

Consider the expansion

$$(1 + x)^8 = {}^8C_0 + {}^8C_1 x + {}^8C_2 x^2 + \dots + {}^8C_8 x^8$$

$$(1 + x)^{17} = {}^{17}C_0 + {}^{17}C_1 x + {}^{17}C_2 x^2 + \dots + {}^{17}C_{16} x^{16} + {}^{17}C_{17} x^{17}$$

$$\Rightarrow {}^8C_0 {}^{17}C_{16} + {}^8C_1 {}^{17}C_{15} \dots + {}^8C_8 {}^{17}C_8$$

$$= \text{coefficient of } x^{16} \text{ in } (1 + x)^{25}$$

$$= {}^{25}C_{16} = {}^{25}C_9$$

7. Ans (6.00)

$$\text{Number of diagonals} = \frac{n(n-3)}{2} = 54$$

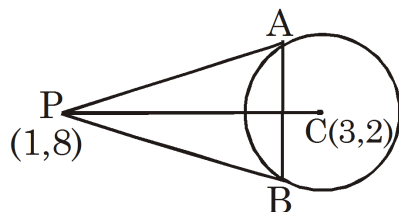
$$\therefore \text{Number of sides } n = 12$$

Number of triangle with exactly one side coinciding with side of polygon is

$$T = 12 \times {}^8C_1 = 96$$

$$\therefore \frac{T}{16} = 6$$

8. Ans (18.25)



Equation of circle circumscribing ΔPAB is

$$(x-1)(x-3) + (y-8)(y-2) = 0$$

$$x^2 + y^2 - 4x - 10y + 19 = 0 \quad \dots(i)$$

Equation of circle passing through points of intersection of circles

$$x^2 + y^2 - 2x - 6y + 6 = 0 \quad \text{and}$$

$$x^2 + y^2 + 2x - 6y + 6 = 0 \quad \text{is given by}$$

$$(x^2 + y^2 - 2x - 6y + 6) + (x^2 + y^2 + 2x - 6y + 6) = 0$$

$$x^2 + y^2 + \frac{(2\lambda - 2)}{\lambda + 1}x - 6y + 6 = 0 \quad \dots(ii)$$

As circle (ii) is orthogonal to circle (i)

So

$$-2 \left(\frac{2\lambda - 2}{\lambda + 1} \right) - 5(-6) = 19 + 6$$

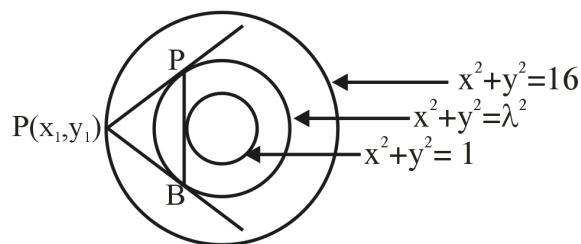
$$\lambda = -9$$

Hence Required equation of circle is

$$x^2 + y^2 + \frac{5}{2}x - 6y + 6 = 0$$

$$R = \sqrt{\frac{25}{16} + 9 - 6} = \frac{\sqrt{73}}{4}$$

9. Ans (2.00)



Let (x_1, y_1) lies on $x^2 + y^2 = 16$

$$\text{So, } x_1^2 + y_1^2 = 16 \quad \dots(i)$$

Now chord of contact of $x^2 + y^2 = 2$

w.r.t $p(x_1, y_1)$

$$xx_1 + yy_1 - \lambda^2 = 0$$

$$xx_1 + yy_1 - \lambda^2 = 0 \quad \text{This is also tangent to circle } x^2 + y^2 = 1$$

$$P_{\perp} = r$$

Length of the perpendicular = Radius of that circle

$$\left| \frac{\lambda^2}{\sqrt{x_1^2 + y_1^2}} \right| = 1$$

$$\left| \frac{\lambda^2}{4} \right| = 1 \Rightarrow \lambda^2 = 4$$

$$\lambda = 2$$

10. Ans (0.00)

$$\frac{100!}{(50!)^2} = 2^{r_1} 3^{r_2} 5^{r_3}$$

Calculate r_1 & r_3